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University of Information Technology & Sciences

Lab Report

Course Title: Internet of Things Lab
Course Code: CSE 402

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Experiment no: 04
Experiment Name: Detection of the light using photoresistor (LDR)

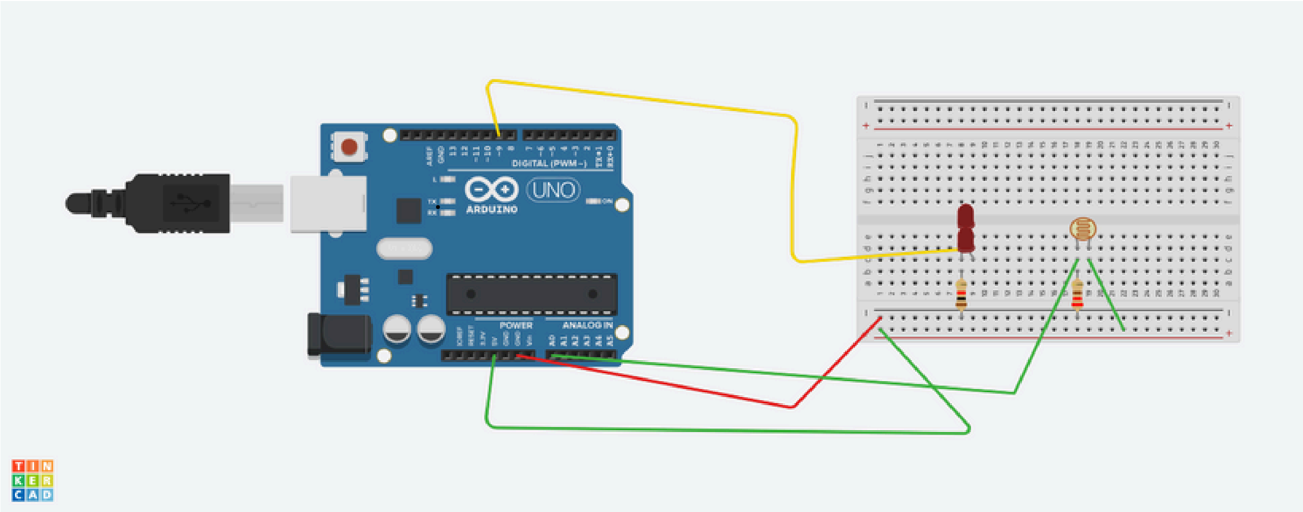
Introduction

In this project I am going to demonstrate how to Control LED through LDR.
This project will involve using a little a bit of code and a very simple circuit that's great for beginners.The video further down this page will go through all the steps to completing this very cool Arduino DIY LED Control with LDR Sensor (Photoresistor)
This project aims to show how to control appliances when darkness appear like smart bulb.
This will serve as a base to build more amazing projects.In every electronic project we must think about what the input and output devices to use, the LDR is one of the best known and cheapest, we use the analog input reading due to the interaction of the project with the environment, along with this We must give importance to the output or presentation of the value

Hardware Required:

Name	Number
1.Breadboard-Small	1
2. AurduinoUno-R3	1
3. Red LED	1
4.Resistor-1KΩ	1
5.JumperWires	several
6. USBPowerCable	1
7.220Ω Resistor	1

Circuit Diagram:



Working Procedure:

1. Insert the photo resistor into your breadboard and connect its pin to the analog pin A0 and the remaining pin to supply on the breadboard.

2. Insert the LED into the breadboard. Attach the positive leg (the longer leg) to pin 9 of the Arduino via the 220-ohm resistor, and the negative leg to GND.

3. Insert the 10K-ohm resistor

4. Upload the code

5. Turn the photo resistor to ON the LED

6. Observe the changes in the state of the LED.

The Sketch :

This sketch works by setting pin A0 as for the photo sensor and pin 9 as an OUTPUT to power the LED. After that the run a loop that continually reads the value from the photo resistor and sends that value as voltage to the LED. The LED will vary accordingly.

Code:

```
const int LDRPin = A0;
const int LEDPin = 9;

const int lightThreshold = 200;

void setup() {

 Serial.begin(9600);

  pinMode(LEDPin, OUTPUT);
}

void loop() {
  int lightLevel = analogRead(LDRPin);

  Serial.print("Light Level: ");
  Serial.println(lightLevel);

  if (lightLevel < lightThreshold) {

    digitalWrite(LEDPin, HIGH);
  } else {

    digitalWrite(LEDPin, LOW);
  }


  delay(200);
}
```

Conclusion:

This project successfully demonstrated light-controlled LED operation using an LDR and Arduino, illustrating basic sensor principles. This foundation enables development of more complex light-sensitive systems, leveraging affordable components for practical applications.